

1.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo... Edit... Save...

Filter: Choose None Apply Stop

Current relation:
Relation: Mall_Customers
Instances: 10

Attributes:
All None Invert Pattern

No. Name

1 ☒ CustomerID
2 ☐ Gender
3 ☐ Age
4 ☐ AnnualIncome
5 ☐ SpendingScore

Attributes: 5
Sum of weights: 10

Selected attribute:
Name: CustomerID
Missing: 0 (0%)
Distinct: 10
Type: Numeric
Unique: 10 (100%)

Statistic	Value
Minimum	1
Maximum	10
Mean	5.5
StdDev	3.028

Class: SpendingScore (Num) Visualize All

Status: OK Log

Windows taskbar: Search, 100%, 24.08.2024, 09:10

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Chosen: None

Current relation: Relation: Mall_Customers-weka.filters.unsupervised.attribute.Remove-RI
Instances: 10

Attributes: 4 Sum of weights: 10

Attributes: All None Insert Pattern

No. Name

- Gender
- Age
- AnnualIncome
- SpendingScore

Remove

Selected attribute: Name: Gender Missing: 0 (0%)

No.	Label	Distinct: 2	Count	Type: Nominal Unique: 0 (0%)	Weight
1	Male		3		
2	Female		7		

Class: SpendingScore (Nom)

Visualize All

Status: OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Chosen: SimpleKMeans with 100 candidates 100 periods: pruning 10000 min-densify 2.0 -H -1.25 -Q -1.0 -N 5 -A "weka.clusterEvaluation.R" -trial fast - 500 min-data 1 1 10

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Cross to cluster evaluation
- ☒ Score clusters for visualization

Ignore attributes: Start Stop

Result list (right-click for options):

- 0/3400: SpendingMeans

Cluster output:

Initial starting points (random):

- Cluster 0: Female, 23, 14, 77
- Cluster 1: Female, 39, 29, 6
- Cluster 2: Female, 26, 16, 6
- Cluster 3: Female, 22, 17, 76
- Cluster 4: Female, 31, 17, 60

Mixing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster# 0	1	2	3	4
	(10.0)	(1.0)	(1.0)	(1.0)	(4.0)	(3.0)
Gender	Female	Male	Male	Male	Female	Female
Age	29.6	21	34	19	26.9	20.6667
AnnualIncome	17	15	19	19	17.5	17
SpendingScore	45.4	81	3	29	79.75	17.3333

Time taken to build model (full training data): 0 seconds

=== Model and evaluation on training set ===

Clustered instances:

	0	1	2	3	4
0	1 (10%)				
1	1 (10%)				
2	1 (10%)				
3	4 (40%)				
4	3 (30%)				

Status: OK

Log

2.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit Save

Filter: Choose None Apply Stop

Current relation: Relation: Employee Instances: 10

Attributes: 5 Sum of weights: 10

Selected attribute: Name: CustomerID Missing: 0 (0%)

Statistics: District: 10 Type: Numeric Unique: 10 (100%)

Statistic	Value
Minimum	1
Maximum	10
Mean	5.5
StdDev	3.028

Class: SpendingScore (Num) Visualize All

Status: OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose SimpleKMeans -init 0 -max-conditions 100 -periods -out 1000 -min-entropy 2.0 -it -125 -c 2 -l 0 -q 5 -A "weka.core.EuclideanDistance -R first-last" -S 50 -num-clusters 1 -S 10

Cluster mode: ☒ Use training set ☐ Sequential test set ☐ Percentage split ☐ Classify to clusters evaluation (cluster SpendingScore) ☒ Store clusters for visualization

Ignore attributes: Start Stop

Result list (right-click for options): 00.000 - SimpleKMeans

Cluster output:

Within clusters sum of squared errors: 0.546030036135034

Initial starting points (random):

Cluster 0: 23,16,77
Cluster 1: 79,18,6
Cluster 2: 20,16,6
Cluster 3: 20,17,76
Cluster 4: 31,17,60

Missing values globally replaced with mean/mode

Final cluster centroids:

	Full Data	0	1	2	3	4
AnnualIncome	(10,0)	(2,0)	(2,0)	(2,0)	(2,0)	(2,0)
Age	28.6	20	49.5	19.5	25	31
AnnualIncome	17	15.5	18.5	15.5	18	17
SpendingScore	49.4	79	4.5	22.5	90.6667	49

Time taken to build model (full training data) : 0 seconds

==== Model and evaluation on training set ====

Clustered Instances:

	0	1	2	3	4
0	2 (20%)				
1	2 (20%)				
2	2 (20%)				
3	3 (30%)				
4	1 (10%)				

Status: OK Log

3.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit Save

File: Choose **None** Apply Cancel

Current relation: student_performance
Instances: 20
Attributes: 4
Sum of weights: 20

Attributes: All None Inverted Pattern

No. Name

- 1 StudyHours
- 2 Attendance
- 3 Assignment
- 4 Result

Remove

Selected attribute: Name StudyHours
Missing: 0 (0%)

No.	Label	Distinct	Count	Type	Missing	Weight
1	Low	7	7	Nominal	7	
2	Medium	6	6	Nominal	6	
3	High	7	7	Nominal	7	

Class Result (Name)

Visualize All

Status: OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:

- ☐ Use training set
- ☐ Split test set
- ☒ Cross-validation: folds: 10
- ☐ Percentage split: % 80
- More options...

(Name) Result: Start Stop

Result for single-click for options: **06.6216 - LogProbabilities**

Classifier output:

	0.0	0.0
Median	10.0	16.0

Assignment:

	0.0	0.0
Incomplete	1.0	0.0
Complete	2.0	10.0
Total	3.0	10.0

Time taken to build model: 0 seconds

==== Cross-validated cross-validation ====

==== Summary ====

	0.0	0.0
Currently Classified Instances	10	10
Incorrectly Classified Instances	5	10
Mean absolute error	0.499	
Root mean squared error	0.707	
Relative absolute error	59.364	%
Root relative squared error	69.162	%
Total Number of Instances	20	

==== Detailed Accuracy By Class ====

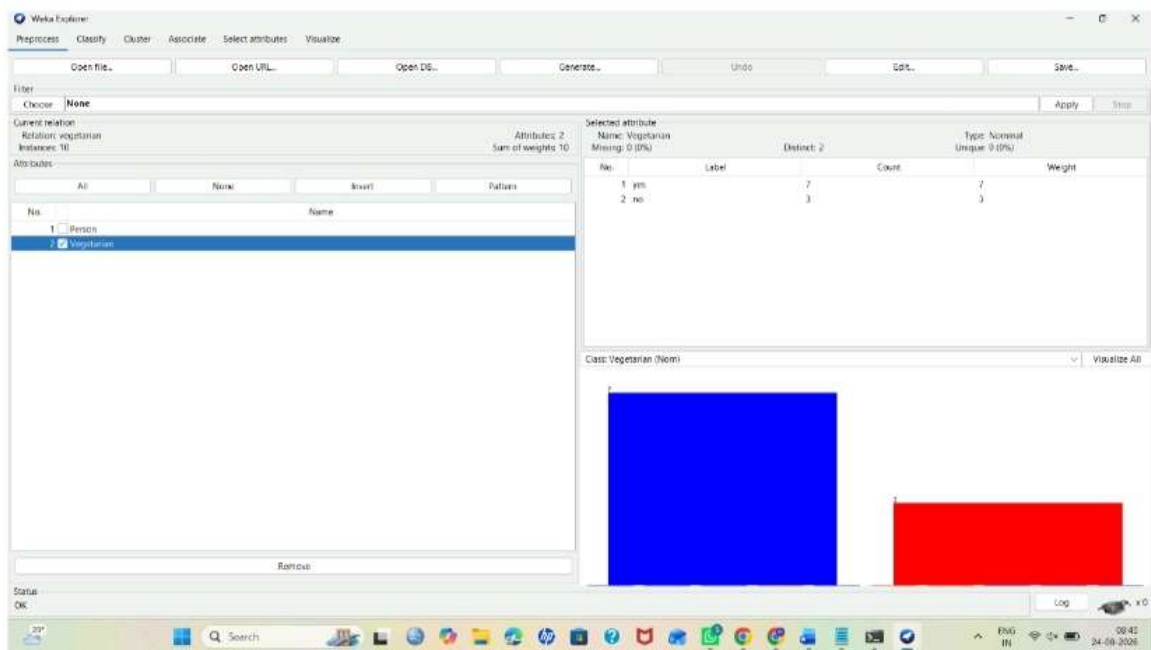
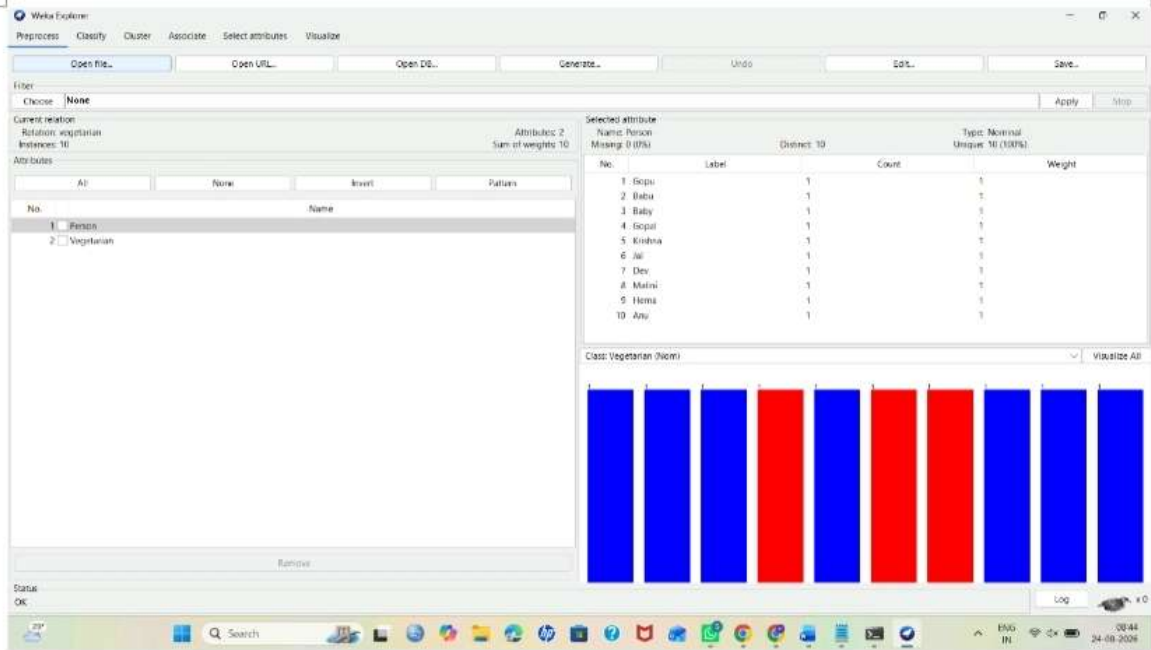
	TP Data	FP Data	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	ROC Area	Class
Weighted Avg.	0.750	0.246	0.750	0.750	0.750	0.750	0.471	0.929	0.929	Pass

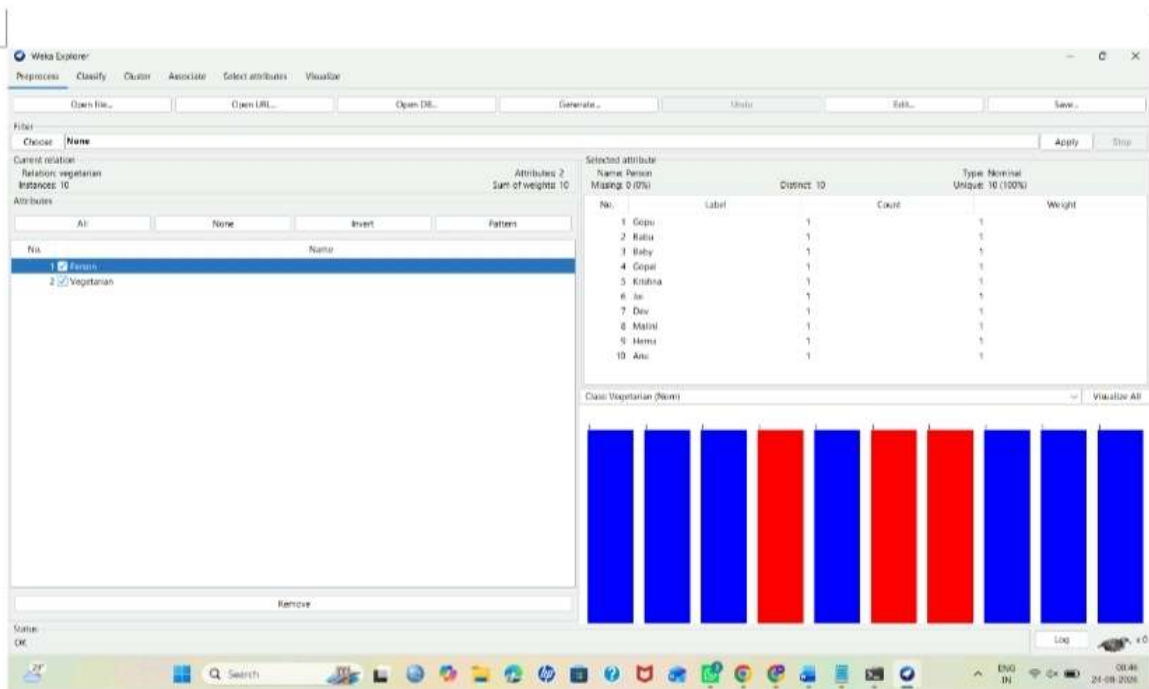
==== Confusion matrix ====

a = 0 -- classified as
0 1 : a = Fail
0 10 : b = Pass

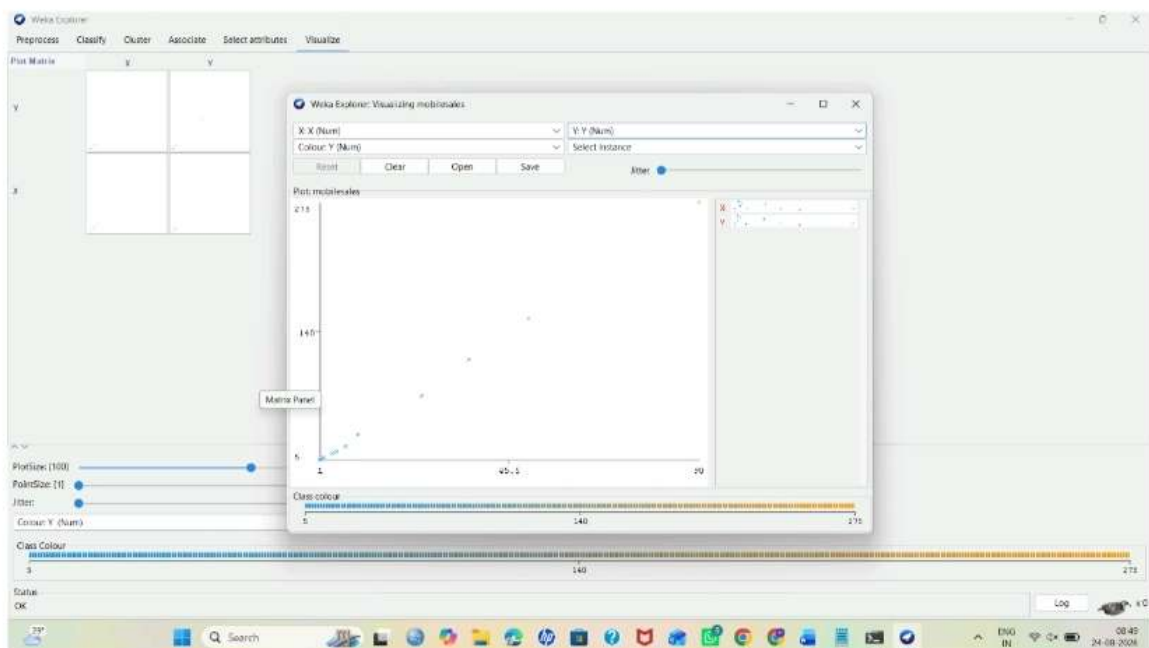
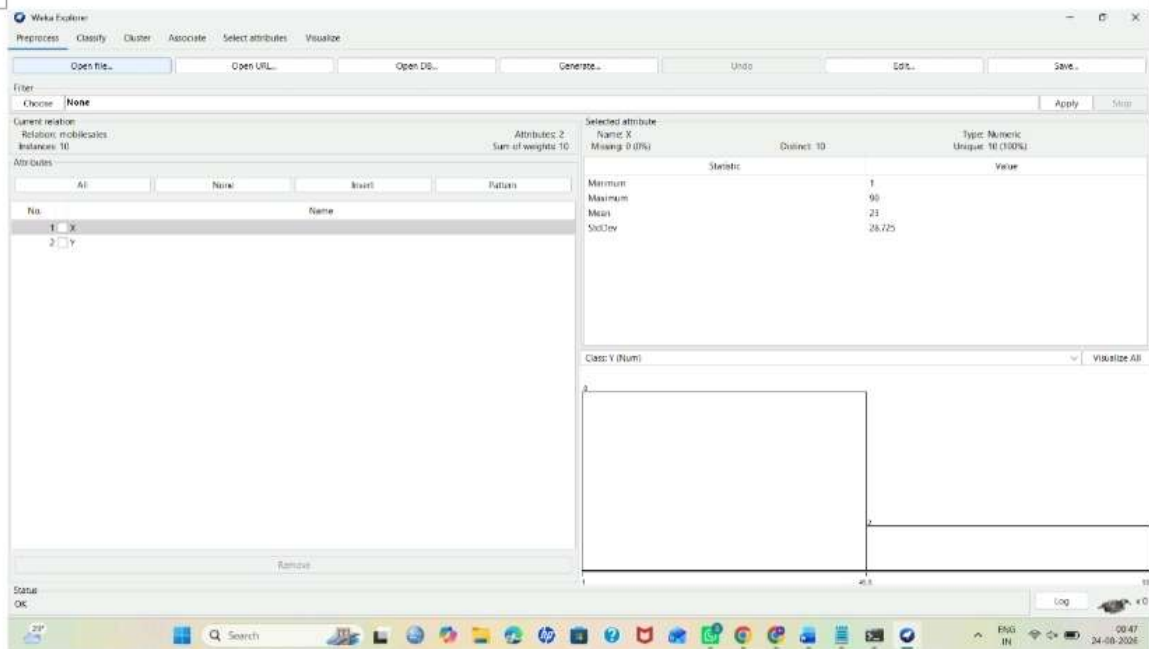
Status: OK Log

4.

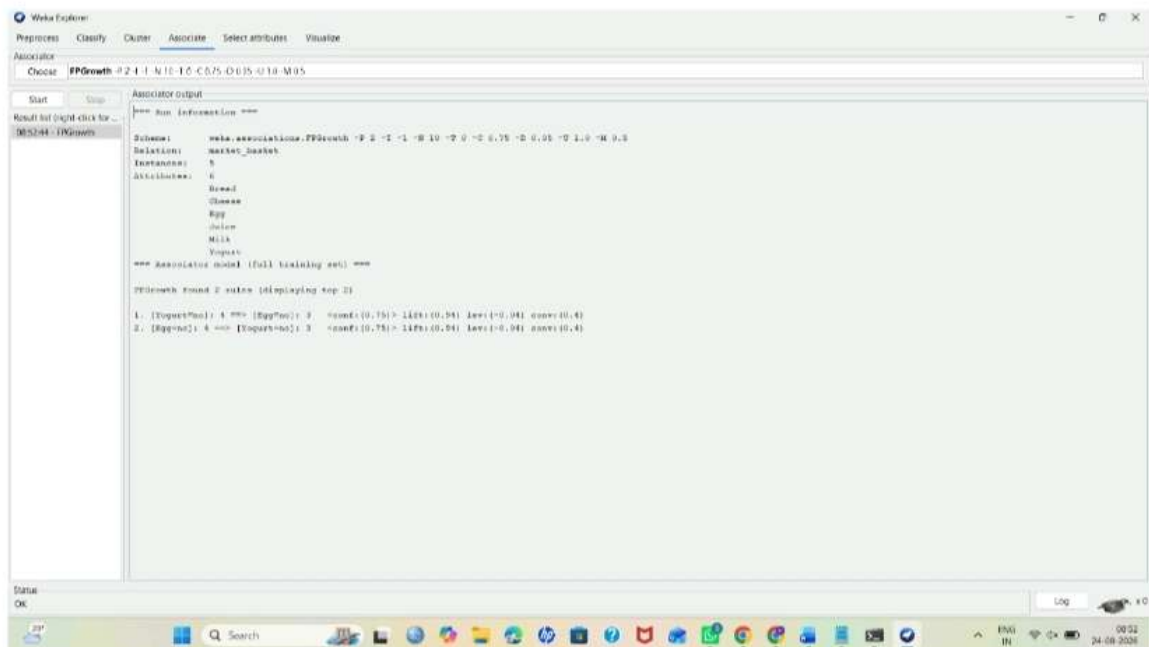
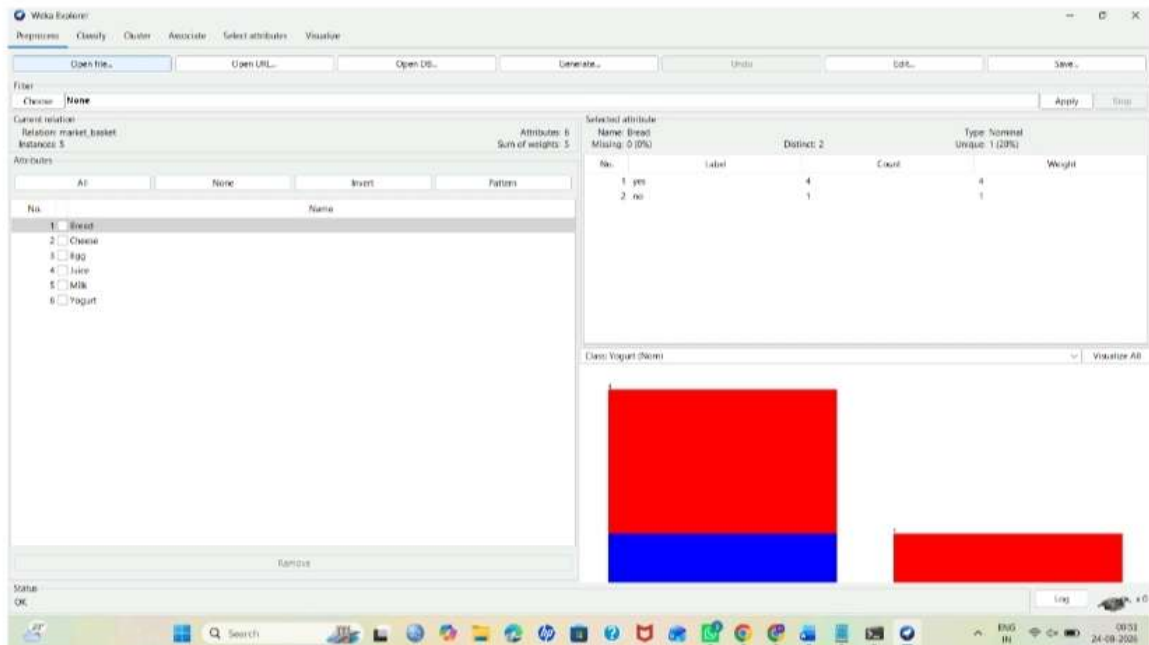




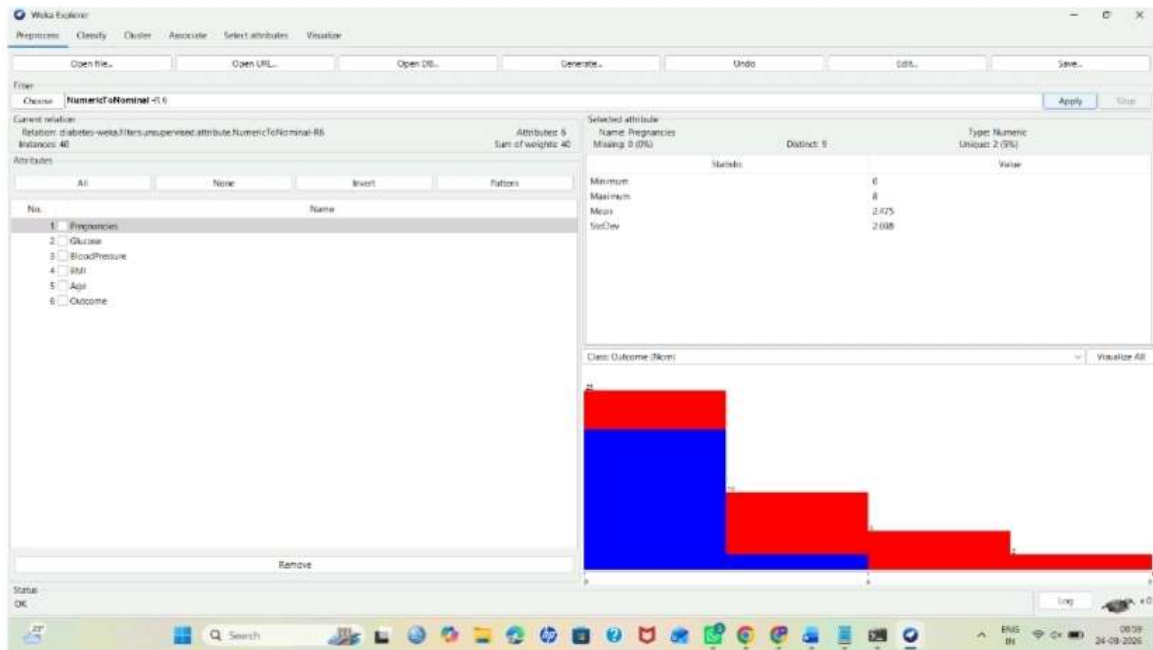
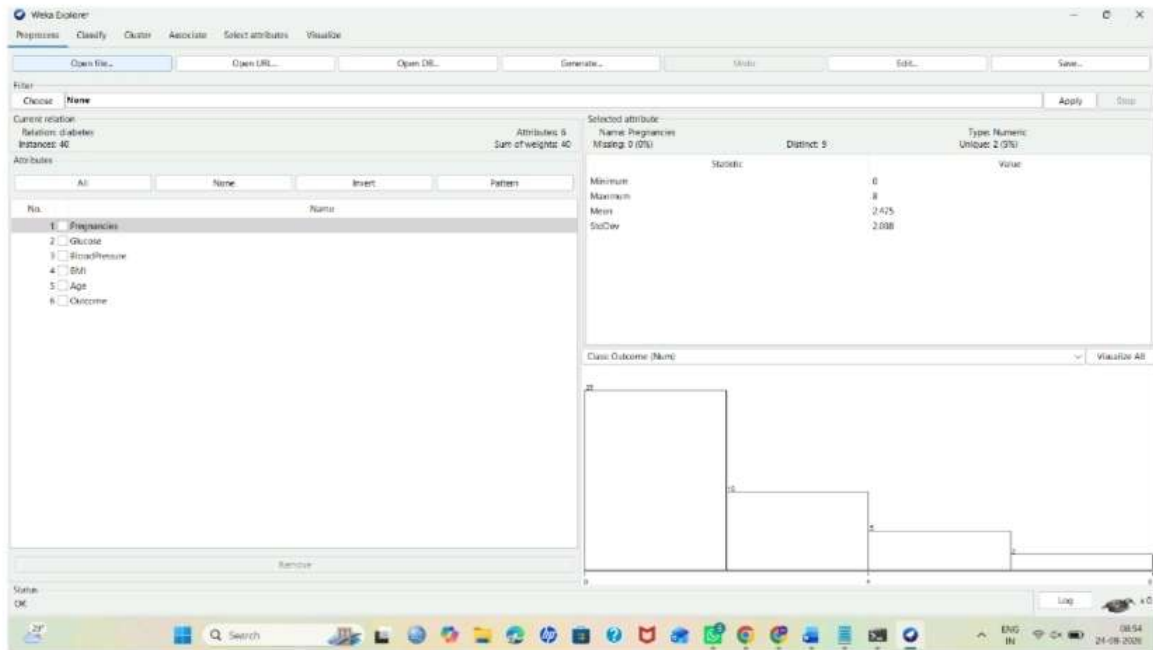
5.

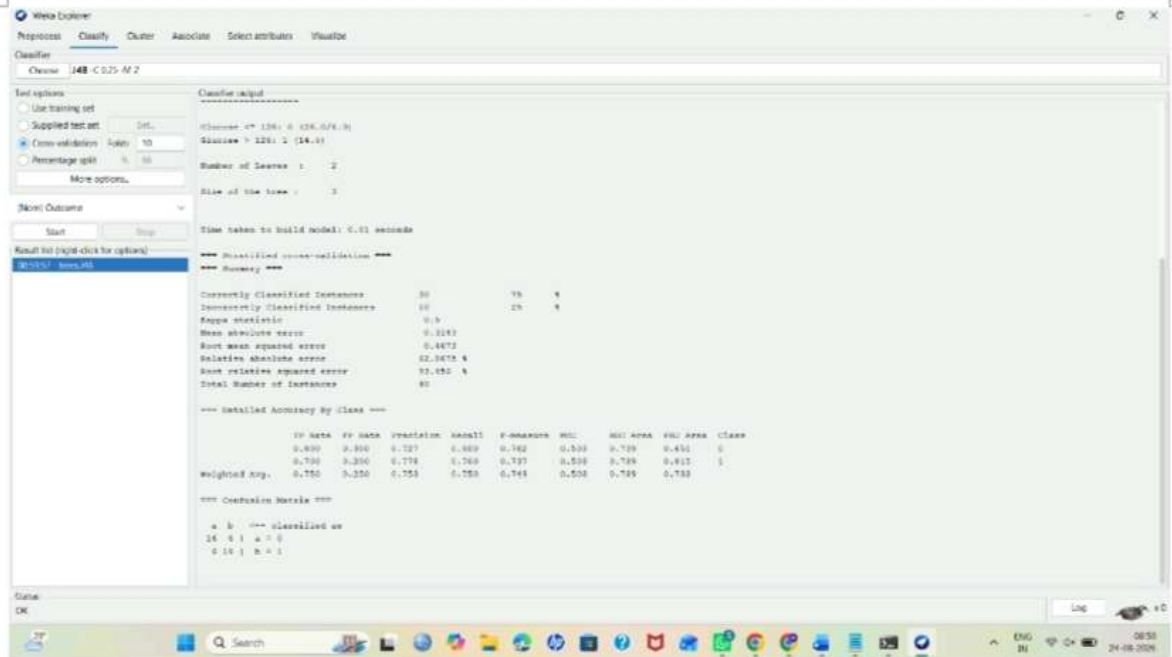


6.

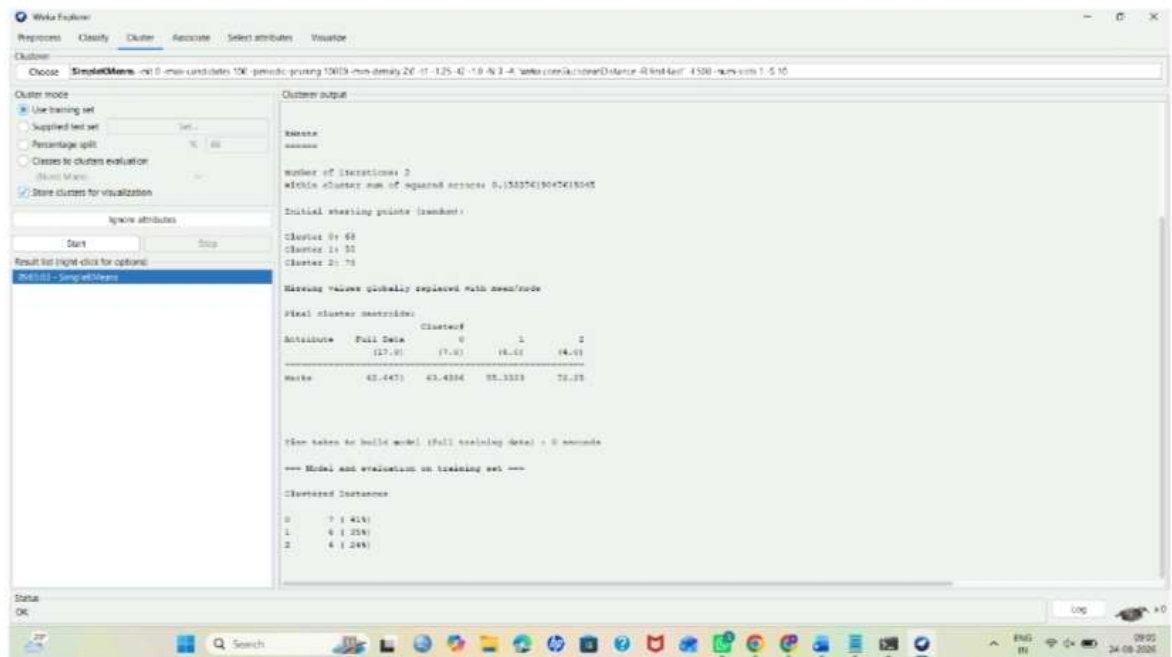


7.

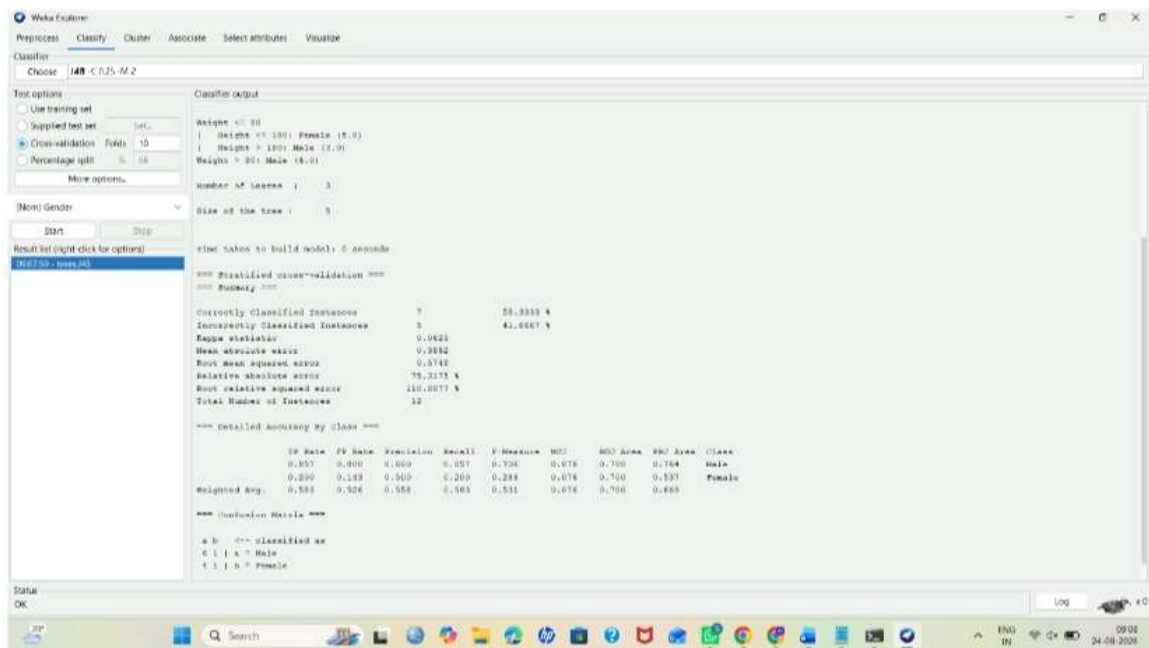
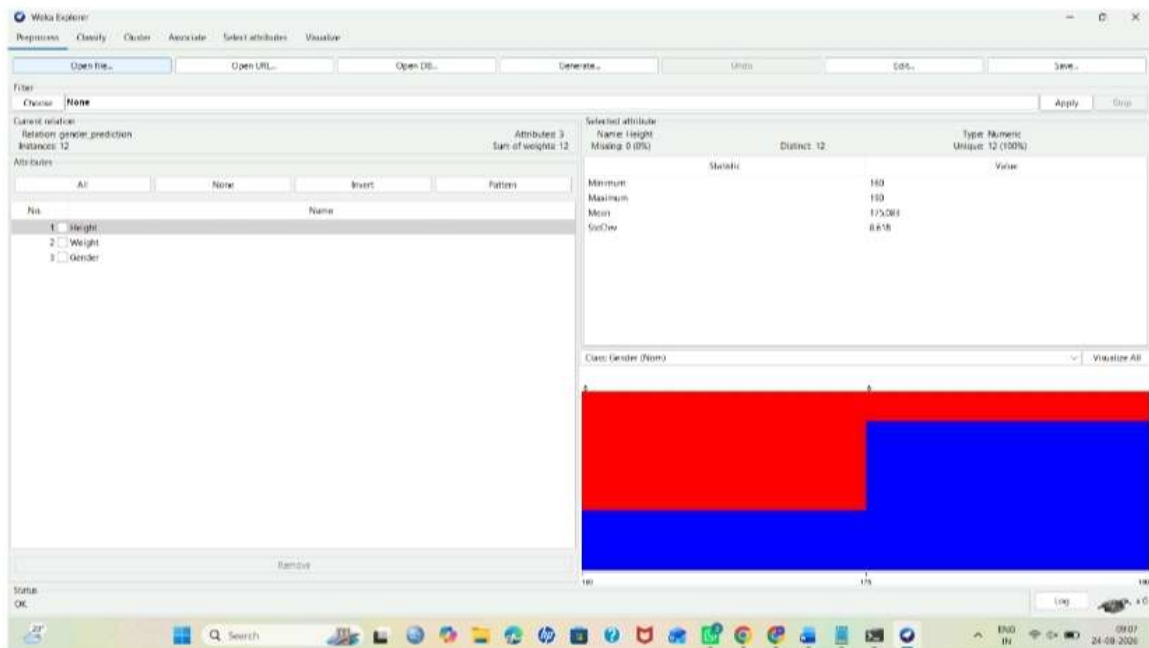




8.



9.



10.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose: None Apply Stop

Current relation: Relations: occupying Instances: 9 Attributes: 5 Sum of weights: 9

Attributes: All None Invert Pattern

No. Name

1 ☒ SONY
2 ☐ BPN
3 ☐ LG
4 ☐ SAMSUNG
5 ☐ ONIDA

Remove

Status: OK

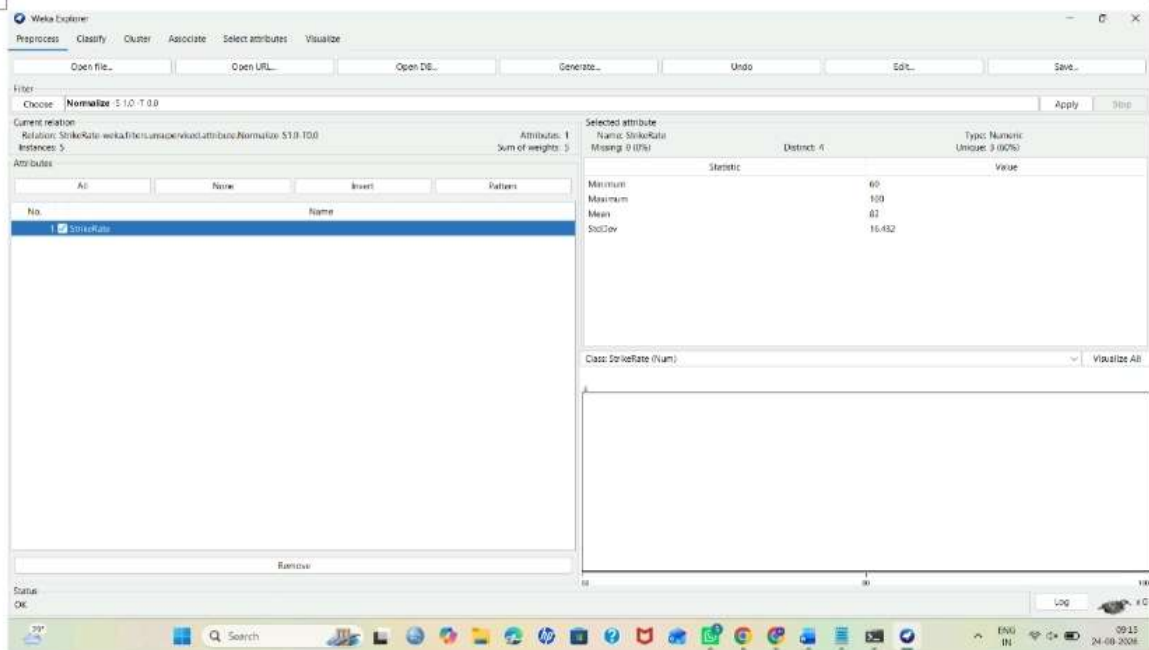
Selected attribute: Name: SONY Missing: 0 (0%) Distinct: 2 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	yes	6	6
2	no	3	3

Class: ONIDA (Nom) Visualize All

Log

24-09-2006 09:09



12.

```

Code Blame 26 lines (23 loc) - 1.43 KB
1 DAY 4 - QUESTION 12: Standard Deviation & Variance Calculation
2 =====
3 Dataset (9 Cars):
4 AvgSpeed (X): [78, 81, 82, 74, 83, 82, 77, 80, 70]
5 TotalTime (Y): [39, 37, 36, 42, 35, 36, 40, 38, 46]
6 n = 9
7
8 1. Mean Calculations:
9 - Mean(AvgSpeed) = (78+81+82+74+83+82+77+80+70) / 9 = 79.77777777777777 ≈ 79.78 kph
10 - Mean(TotalTime) = (39+37+36+42+35+36+40+38+46) / 9 = 39.77777777777778 ≈ 39.78 mins
11
12 2. (a) & (b) Variance and Standard Deviation for AvgSpeed:
13 - Sum of squared differences from mean:
14 (78-79.78)^2 + (81-79.78)^2 + (82-79.78)^2 + (74-79.78)^2 + (83-79.78)^2 + (82-79.78)^2 + (77-79.78)^2 + (80-79.78)^2 + (70-79.78)^2
15 = 0.31 + 5.95 + 11.83 + 20.79 + 19.71 + 11.83 + 2.43 + 2.87 + 73.27 = 148.22
16 - Sample Variance (s^2) = 148.22 / (9 - 1) = 18.53
17 - Sample Standard Deviation (s) = sqrt(18.53) ≈ 4.30 kph
18 (Population Variance sigma^2 = 16.47, Population sigma = 4.06 kph)
19
20 3. (a) & (b) Variance and Standard Deviation for TotalTime:
21 - Sum of squared differences from mean:
22 (39-39.78)^2 + (37-39.78)^2 + (36-39.78)^2 + (42-39.78)^2 + (35-39.78)^2 + (36-39.78)^2 + (40-39.78)^2 + (38-39.78)^2 + (46-39.78)^2
23 = 0.85 + 3.17 + 7.73 + 10.37 + 14.29 + 7.73 + 1.49 + 8.61 + 52.13 = 97.56
24 - Sample Variance (s^2) = 97.56 / (9 - 1) = 12.19
25 - Sample Standard Deviation (s) = sqrt(12.19) ≈ 3.49 mins
26 (Population Variance sigma^2 = 10.84, Population sigma = 3.29 mins)

```

13.

```

Code Blame 32 lines (29 loc) · 1.44 KB
1 DAY 4 - QUESTION 13: Meta-Rule Matching & Mining Comparison
2 -----
3 Dataset:
4 T100: (M, O, N, K, E, V)
5 T200: (O, O, N, K, E, V)
6 T300: (M, A, K, E)
7 T400: (M, U, C, K, V)
8 T500: (C, O, K, I, E)
9
10 (a) Apriori vs FP-Growth Mining Process & Efficiency Comparison:
11 - Frequent Itemsets (minsup = 3 / 60%):
12   * 1-itemsets: {K}:5, {E}:4, {M}:3, {O}:3, {V}:3
13   * 2-itemsets: {K, E}:4, {K, M}:3, {K, O}:3, {K, V}:3, {E, O}:3
14   * 3-itemsets: {K, E, O}:3
15 - Efficiency Comparison:
16   * Apriori: Scans entire database multiple times (per candidate length k) and incurs large candidate generation and testing overhead.
17   * FP-Growth: Scans DB only twice, constructs a compact FP-Tree in memory, and uses divide-and-conquer conditional pattern trees, being significantly faster
18
19 (b) Strong Association Rules matching Meta-Rule:
20 Meta-Rule Template: buys(X, item1) ^ buys(X, item2) => buys(X, item3)
21 [Form: {item1, item2} -> {item3}]
22
23 Extracted from Frequent 3-Itemset {K, E, O} (Support = 3/5 = 60%):
24 1. {K, O} -> {E}:
25   - Support = 3/5 = 60%
26   - Confidence = Support({K, E, O}) / Support({K, O}) = 3/3 = 100.0%
27 2. {E, O} -> {K}:
28   - Support = 3/5 = 60%
29   - Confidence = Support({K, E, O}) / Support({E, O}) = 3/3 = 100.0%
30 3. {K, E} -> {O}:
31   - Support = 3/5 = 60%
32   - Confidence = Support({K, E, O}) / Support({K, E}) = 3/4 = 75.0%

```